9 Cryptography

9.1 Intro to Cryptography

In this chapter, we will be assisting Alice and Bob communicate with each other. They are very secretive and do not like their conversations to be known by others.
Then there's Eve...

Let's talk about how we might do this for a second. Suppose that you and the person next to you live in different cities (for the purpose of this exercise, we all live in a time where messages must be written down and taken by a courier to wherever they need to go). You are visiting your friend and you decide you want to be able to send secret messages to each other, but you don't trust your couriers not to read your message. How might you decide to make your messages secret?

9.2 Shift Ciphers

Example 9.2.1 A shift cipher is a simple kind of cipher where the alphabet is shifted to obscure the original message. For example, if we choose to use a shift cipher, and our secret key is 3, then we encode messages according to the following:

```
ABCDEFGHIJKLMNOPQRSTUVWXYZ
   ABCDEFGHIJKLMNOPQRSTUVWXYZ
```

Example 9.2.2 Use a shift cipher with key 5 to encrypt the message "We ride at dawn".

```
ABCDEFGHIJKLMNOPQRSTUVWXYZ
     ABCDEFGHIJKLMNOPQRSTUVWXYZ
```

Example 9.2.3 The following message was sent to you by a friend, and you know that they used a shift cipher with key 8. What was the original message?

```
LWVBFMIBFBPMFNQAP
```

How secure is this method? What should we consider a 'secure' method?

knowing what encryption scheme is being used should not make it easier to decrypt the message